

V2B Synthetic Data: Users, Cars, Sessions

How three CSVs encode behavior, physics, and arrivals

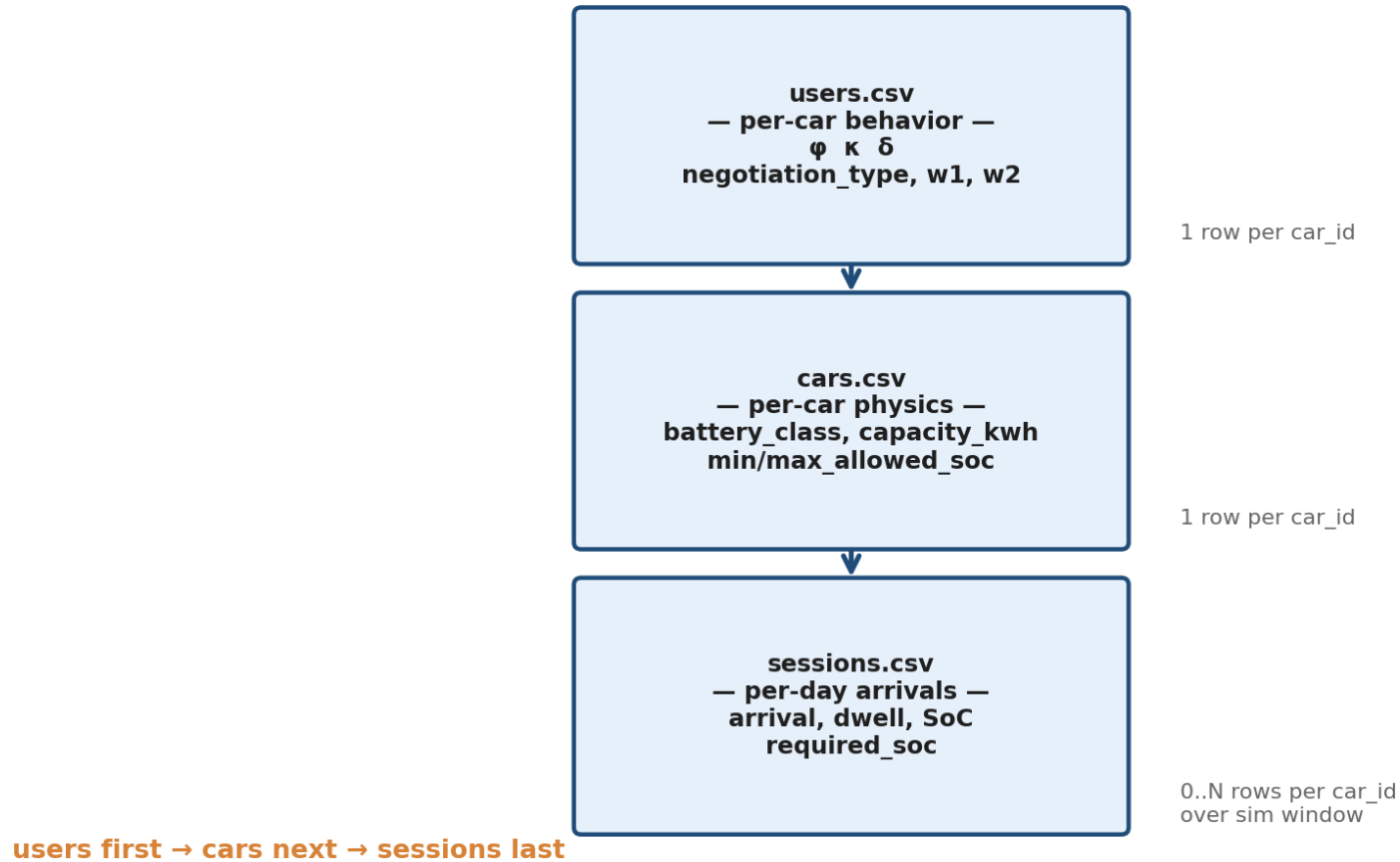
Rishav Sen — Vanderbilt — 2026-05-20

Short overview — 10 slides.

For full detail: `OVERVIEW.md`, per-CSV appendix slides, `sessions.py`.

The three CSVs

The three CSVs — generated in order



Generated in order: users → cars → sessions. Each car_id gets exactly one users.csv + one cars.csv row. Each car_id generates 0..N sessions across the sim window.

The three behavioral axes

φ — phi (frequency)

P(show up on any given day). $\varphi \in [0, 1]$, $\varphi \sim U(\text{freq_lo}, \text{hi})$.

Bites: Bernoulli gate, sessions step 2. $\varphi = 0.84 \Rightarrow \sim 84\%$ of weekdays.

κ — kappa (consistency)

How regular arrival timing is. $\kappa \in [0, 1]$, $\kappa \sim U(\text{consist_lo}, \text{hi})$.

Bites: $\sigma_{\text{eff}} = \sigma \cdot (1 - \kappa \cdot 0.5)$. High $\kappa \Rightarrow$ tight; low $\kappa \Rightarrow$ spread.

δ — delta_km (commute)

One-way commute distance. $\delta \in [3, 100]$ km, $\delta \sim U(\text{dist_lo}, \text{hi})$.

Bites: $\text{shift_eff} = \text{shift} - 0.003 \cdot \delta$. Longer commute $\Rightarrow$ lower arrival SoC.

Region grid (5 default regions)

Each car is assigned one region, which fixes its (φ , κ , δ) bounds.

Region	φ range	κ range	δ range (km)	Weight	Mental model
stable_commuter	[0.85, 1.00]	[0.75, 1.00]	[40, 80]	0.35	Long-distance office, daily
flexible_local	[0.70, 0.95]	[0.50, 0.80]	[5, 15]	0.25	Local, frequent + flexible
irregular_distant	[0.40, 0.70]	[0.20, 0.50]	[40, 100]	0.20	Long commute, ~3 days/wk
occasional_visitor	[0.05, 0.20]	[0.10, 0.40]	[3, 50]	0.10	Rare drop-in
erratic	[0.30, 0.70]	[0.05, 0.30]	[5, 80]	0.10	Unpredictable schedule

Region weights themselves can be Dirichlet-perturbed per sample (knob: `axes_distribution_dirichlet_alpha`).

users.csv generation

For each `car_id` $\in [1, \text{ev_count}]$:

1. **(opt.) Dirichlet perturb region weights** — if $\alpha < 1e6$, `realized_weights` $\sim \text{Dirichlet}(\text{weights} \cdot \alpha)$.
2. **Assign region** — `region` $\sim \text{Categorical}(\text{realized_weights})$.
3. **Sample axes** — $\phi \sim U(\text{freq_lo}, \text{hi})$, $\kappa \sim U(\text{consist_lo}, \text{hi})$, $\delta \sim U(\text{dist_km_lo}, \text{hi})$.
4. **Negotiation type** — `Categorical(negotiation_mix)`; 4 CONSENT clusters (I/II/III/IV).
5. **CONSENT weights** — $(w1, w2) \sim N(\text{cluster_}\mu, \text{cluster_}\sigma)$, clipped ≥ 0 , $\times w_multiplier$. *(used for downstream negotiation)*

Schema: `car_id, region, phi, kappa, delta_km, negotiation_type, w1, w2`

`cars.csv` generation

Branch on `battery_heterogeneity`:

homog

all cars =
`argmax(battery_mix)` .
Dirichlet ignored.

mixed, $\alpha \geq 1e6$ (default)

per car: `battery_class ~`
`Categorical(battery_mix)`

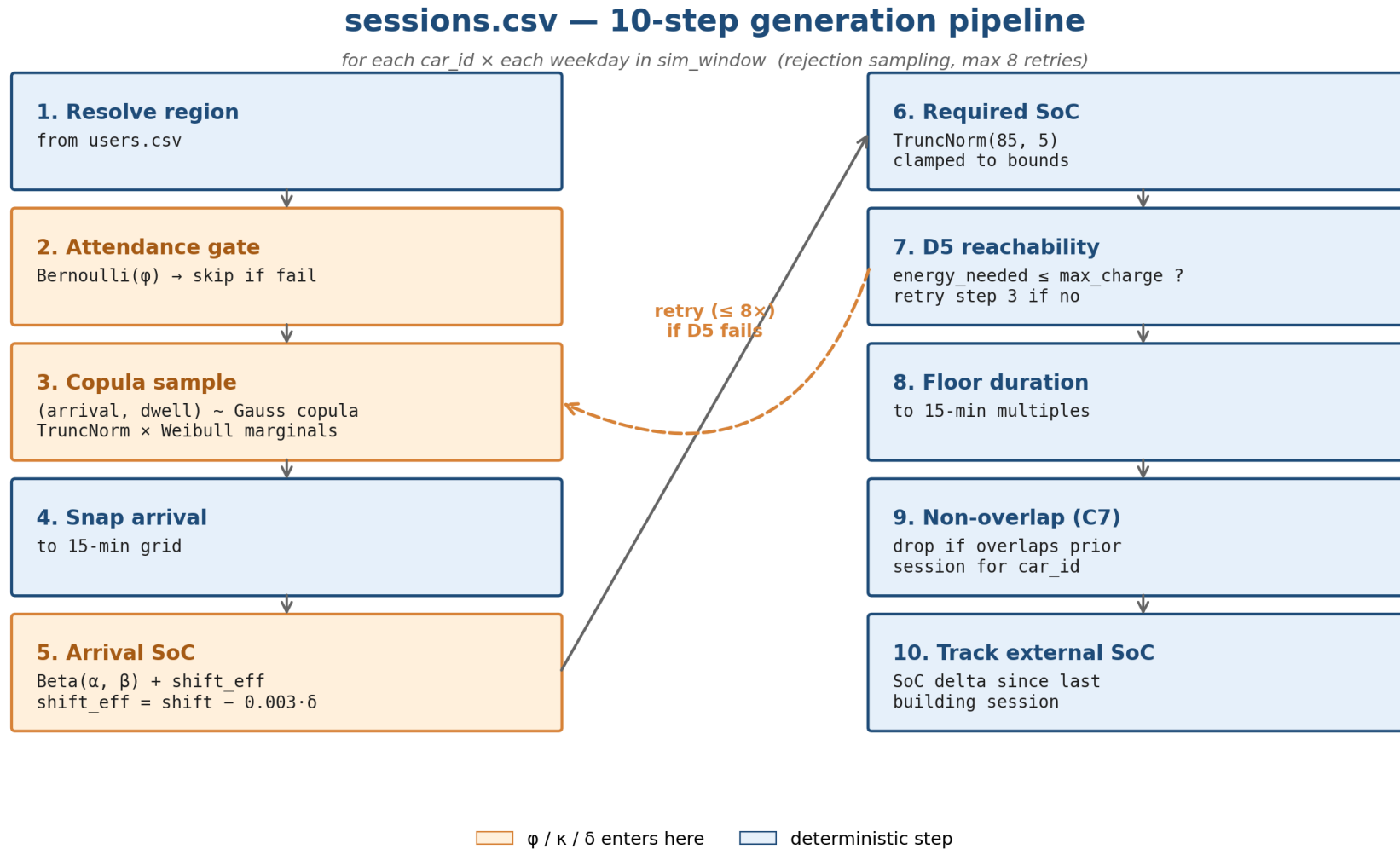
mixed, $\alpha < 1e6$

once/sample: `realized_mix`
`~ Dirichlet(battery_mix .`
 `$\alpha$ )` ; per car:
`Categorical(realized_mix)`

Then lookup `BATTERY_SPECS[battery_class]` $\rightarrow$ `capacity_kwh` $\in \{24, 40, 75, 100\}$
and SoC bounds (typ. `[10%, 100%]`).

Schema: `car_id`, `capacity_kwh`, `min_allowed_soc`, `max_allowed_soc`,
`battery_class`

sessions.csv pipeline (10 steps)

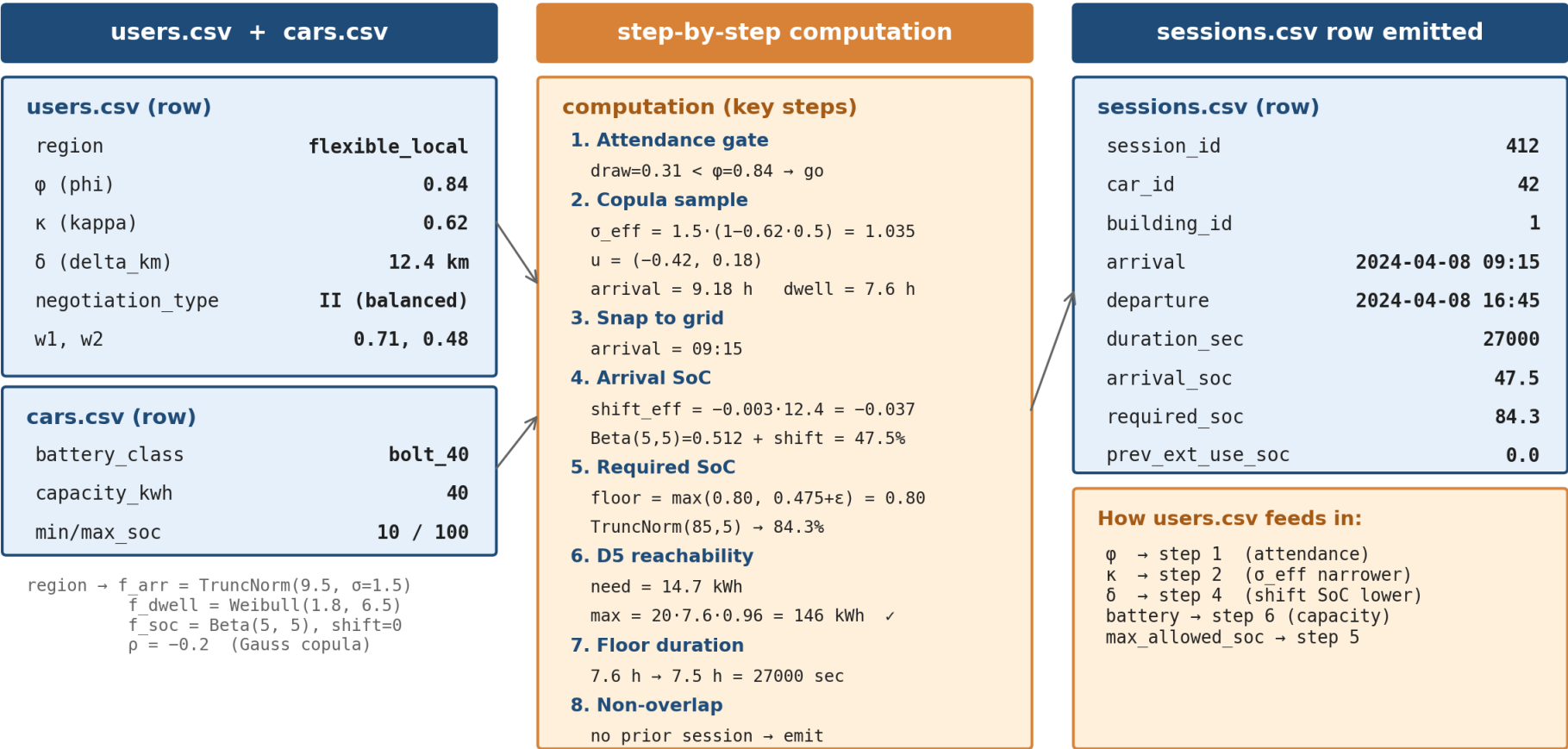


Step 3 model: bivariate **Gaussian copula** on (arrival_hour, dwell_hours) — marginals $\text{TruncNorm}(\mu, \sigma) \times \text{Weibull}(k, \lambda)$: ϕ per region (typ. negative ⇒ early arrivers stay longer). **Rejection sampling: D5-fail car-days retry ≤ 7 8x then drop.**

v2b_syndata ← users, cars, sessions snap overview

Worked example

Worked example — car_id = 42, day = 2024-04-08 (Mon)



car_id=42, day=2024-04-08 — full numerical trace; same numbers reproduced in the .docx.

Knob cheat sheet

Bucket	Knob	Effect
ev_fleet	ev_count ; battery_mix ; battery_heterogeneity ; battery_mix_dirichlet_alpha	Fleet size + battery class branch & mix
user_behavior	axes_distribution ; axes_distribution_dirichlet_alpha	5-region grid: (ϕ , κ , δ) bounds + weight perturbation
user_behavior	negotiation_mix ; w_multiplier	CONSENT cluster mix + (w1, w2) scale
user_behavior	min_depart_soc	Floor on required_soc_at_depart
user_behavior	region_distributions.<r>.<dist>.<param>	Deep override of any region's f_arr / f_dwell / f_soc / ρ
charging_infra	charger_count ; *_rate_kw ; directionality_frac	Gate D5 reachability check
sim_window	mode ; weekdays_only ; start ; custom_end	Day loop in sessions.py
noise	arrival_time_jitter_min ; soc_arrival_jitter_pct ; profile	Post-render jitter; tmyx_stochastic $\Rightarrow \pm 5$ min, $\pm 3\%$

Where to go deeper

This deck is the short version. For more:

- `walkthrough.html` — interactive page: drag sliders (φ , κ , δ , ρ , Dirichlet α , CONSENT cluster) and see distributions update live
- `OVERVIEW.md` — full architecture, all 7 CSVs, manifest, descriptor model
- **Per-CSV appendix slides** — every CSV's full pipeline (slides A9–A15 in the long deck)
- `src/v2b_syndata/renderers/sessions.py` — implementation
- `configs/populations.yaml` — per-region distribution parameters
- `KNOB_REFERENCE.md` — every knob, with audit notes

Questions?